

PWAs have so far delivered great results for businesses as they involve less data usage and a quick loading time. They improve the user experience and also user retention rates. They make the app more secure and cost less to develop.

Characteristics of PWAs

The essential features of PWAs are:

- A home screen icon
- Push notifications
- Splash screen
- Geolocation
- Offline mode
- Background synchronisation
- Media access from audio and video devices
- The latest technology
- Secured
- Provide a super-rich experience
- No app store is required

The adoption of PWAs by Pinterest resulted in a 60 per cent hike in its engagements and a 44 per cent increase in advertisement revenue. The time visitors spent on the site also increased by 40 per cent. Tinder reduced its loading time from 11.91 seconds to 4.69 seconds with its new PWA. BookMyShow's PWA has helped it to increase conversion rates by over 80 per cent. Twitter had a 75 per cent increase in tweets sent and a 20 per cent fall in bounce rates when it used a new PWA. Pure Formulas is a leading online health and supplements company based in the US. By introducing PWAs, the company has enjoyed a 14 per cent increase in conversions, a 9 per cent increase in average order value (AOV) and a 23 per cent increase in revenue per visit.

To put it simply, PWAs are Web applications that load in a Web browser just like Web pages or websites. They give users a rich mobile experience via native-like functionalities such as the ability to work offline, push notifications and device hardware accessibility.

Some of the situations in which PWAs are exceedingly useful are:

Native apps vs PWAs

Native apps	PWAs
Require a separate code base for each platform such as Android, iOS, etc.	They are built using the same code for several platforms (such as Android, iOS).
Indoor geolocation is possible.	Indoor geolocation is not possible.
Require a higher budget to write platform-specific code.	Require a lower budget to write code since PWAs need less funds.
Need to be downloaded from the app store.	There's no need to download from the app store.
Require more space.	Require less space.
Have a higher data consumption rate and need a faster network.	Have a lower data consumption rate and faster installation.
Low conversion rate.	Seamless conversion.

- The increased app size
- Infrequent use of the app
- Low Internet connectivity
- Big Data and storage consumption
- Frequent updates and permissions
- The number of alternatives for a particular purpose, such as food delivery, e-commerce and so on

Tools to develop PWAs

PWAs will replace 50 per cent of native apps by 2020. Today, thousands of companies around the world are building PWAs. Every company wants to jump into PWA development to be able to deliver better services. Web standard technologies like JavaScript, HTML and CSS are required to build PWAs.

Early 2010 experienced dynamic Web pages that allowed technologies such as interactive Web applications, responsive Web design and screen-size flexibility. These helped the growth of PWAs to a great extent.

The following tools are very useful in the development of PWAs:

- *Pwa.rocks* offers examples of excellent solutions under several categories, including business, shopping and games.
- Google Developers provides resources that help to code and has a comprehensive PWA checklist. For example, the Lighthouse extension

allows you to audit Web pages to determine if they qualify as PWAs.

- PWA Builder takes data from your website for easy and fast generation of the PWA.
- Knockout is a JavaScript library that helps you to create the display and edit user interfaces.
- Angular and React are excellent JavaScript frameworks for PWA development.

The disadvantages of PWAs

- PWAs consume more battery power compared to native apps because they are written in high-level Web code.
- The features of PWAs are still relatively new to most developers.
- There are no budgets for PWA development within companies. So Web developers should be given opportunities, along with support in terms of money and time, to explore this new technology.
- PWAs can't access various device features. For example, they can't access a device's Near Field Communication (NFC). PWAs also can't access a device's Bluetooth, proximity sensors, ambient light, advanced camera controls, geofencing, wake lock, contacts and more. Because of these reasons, many users are still not very familiar with PWAs.